

A Comparative Percolation Analysis of the Internet AS Topology and the World Wide Web

Work plan for the final project

Modelling Real World Problems (900393SCIY)

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1. Study topic and context

Modern society depends on two large-scale information infrastructures: the Internet at the level of Autonomous Systems, and the World Wide Web at the level of hyperlinked pages (Zhang et al., 2015). Both are commonly described in the network-science literature as scale-free networks, meaning that their degree distributions are dominated by a small number of highly connected hubs (Kawasumi et al., 2024). A central qualitative claim about these networks is that they are simultaneously robust to random failure and fragile under targeted attack (Albert et al., 1999; Zhang et al., 2015). Although this property has been demonstrated theoretically for the Barabási–Albert (BA) model and is supported by classical empirical work, it remains an open question to what extent the two infrastructures behave identically in practice (Albert et al., 1999; Kawasumi et al., 2024). The AS-level Internet and the hyperlink Web, indeed, differ in their growth mechanism, their typical degrees, and higher-order structural features (e.g. clustering) (Faloutsos et al., 1999; Zhou et al., 2004; Kawasumi et al., 2024; Donato et al., 2008). Works such as by Huang et al. (2013) and Zhang et al. (2015) suggest that higher-order structural features, especially clustering, affect the vulnerability of networks, yet, while AS-level Internet and the hyperlink Web have been extensively explored separately, comparative analyses are missing (Kawasumi et al., 2024). A direct, controlled comparison of how the two systems behave under random failure and targeted attack is, therefore, informative both as a test of theory and as input to discussions about infrastructure resilience and cybersecurity policy.

2. Research question

If only a small fraction of nodes could be protected against targeted attack, to what extent would the network resilience be augmented in the AS-level Internet versus the WWW hyperlink network?

In this research, the effect of node removal strategy (random failure and degree-targeted attack) on the size of the giant connected component in the Internet AS-level topology and the WWW hyperlink network will be investigated, and the empirical results will be compared with the predictions of equivalent Erdős–Rényi (ER) and Barabási–Albert (BA) synthetic networks. The project, therefore, addresses the knowledge gap mentioned above by conducting a comparative analysis of the behaviour of the Internet AS-level topology and the WWW hyperlink network under random failure and targeted attacks. Two more specific sub-questions, indeed, follow from the previous general formulation:

- Do both real networks exhibit the asymmetry between random failure and targeted attack predicted for scale-free graphs, and is the asymmetry the same for the two networks?
- Is the heavy-tailed degree distribution alone sufficient to reproduce the observed fragility, or do higher-order structural features (e.g. clustering, degree–degree correlations) contribute?

3. Model description

3.1 Basic model

The basic model consists of a percolation process on an undirected, unweighted graph. After each removal step, the size of the largest connected component, normalized by the original number of nodes, is recorded. Two removal orderings are considered: a random ordering, corresponding to uniform random failure, and a descending-degree ordering, corresponding to a targeted attack on hubs.

Four networks are studied within the same framework:

- **AS-level Internet topology:** the Oregon-1 snapshot from the SNAP repository, where nodes are Autonomous Systems and edges are observed BGP (Border Gateway Protocol) peering relationships.
- **WWW hyperlink network:** the web-NotreDame graph from SNAP, treated as undirected, where nodes are pages from the University of Notre Dame (domain nd.edu) and edges are hyperlinks.
- **ER null model:** an Erdős-Rényi random graph with the same number of nodes and the same average degree as each real network.
- **BA null model:** a Barabási-Albert preferential-attachment graph with parameters chosen to match the average degree of each real network.

3.2 Potential extensions

- **Targeted attack strategy based on betweenness centrality,** as done by other studies (Holme et al., 2002; Crucitti et al., 2004).
- **Configuration model.** Besides the BA and the ER models used as baselines, we could implement a configuration model, a random graph model with a specified degree sequence (Newman, 2018). Unlike the ER and BA models, the configuration model preserves the degree of each vertex of the original network. By comparing the percolation analysis on the configuration model networks with the corresponding real-world network, we would isolate the impact of the degree sequence on the network's robustness and explore the effect of higher-order structures (i.e. beyond the degree sequence), such as clustering, on resilience (Mahadevan et al., 2006; Fosdick et al., 2018).
- **Model with dynamical redistribution of loads, for the AS-level Internet topology.** For networks where loads can redistribute among the neighbouring nodes, targeted attacks can produce a cascade of overload failures (e.g. the damage of a node with a large initial load produces a cascade of overload failures which involve the entire or a substantial part of the network) (Motter et al., 2002; Crucitti et al., 2004). In this model, each node's load corresponds to the betweenness centrality, which will be recalculated after each removal in percolation; node's capacity would be introduced, by using the formula in Crucitti et al. (2004), where the capacity is proportional to the initial load.
- **Directed network model, for the WWW hyperlink network.** While in the basic model the WWW network is treated as undirected for simplicity and comparability, it is inherently a directed network (Liu et al., 2005). We could investigate its robustness when treated as a directed graph; if so, the size of the giant strongly connected component would be measured.
- **Targeted attack strategy based on PageRank centrality, for the WWW hyperlink network.** PageRank centrality takes into account the number and quality of links to a node and could, therefore, be used for targeted attacks in the WWW (directed) hyperlink network. This would allow us to evaluate each page's importance based on the importance of its hyperlinks, not only the number of hyperlinks, and explore the effect of targeted attacks on the most important pages.

4. Data collection and in silico experiments

4.1 Data and Experimental design

All network data are obtained from the Stanford Large Network Dataset Collection (SNAP, <https://snap.stanford.edu/data/>).

We will implement three comparisons, chosen to isolate specific structural contribution to network fragility and to control for confounds caused by differences in network size and density:

1. Within each network, the giant-component response under uniform random removal is compared to the effect of degree-targeted removal. This comparison is unaffected by the empirical differences between the two networks as each network is only compared against itself.
2. Each network is then compared with the ER and BA synthetic networks generated to match its number of nodes and average degree. The ER null provides a baseline in which the heavy-tailed degree distribution is absent, while the BA null reproduces the heavy tail through preferential attachment.
3. The two real networks are compared not in terms of their percolation curves, but in terms of their deviations from their respective BA nulls. Similar systematic deviations across the two systems would indicate a shared higher-order mechanism, whilst divergent deviations would indicate that fragility beyond the scale-free explanation is system-specific.

Further remarks:

- Since percolation is a stochastic process, all experiments will be repeated 10-20 times and average behaviours will be analysed.
- In percolation with degree-targeted removal, degree centrality will be recalculated after each node removal. As shown by Holme et al. (2018), an attack strategy based on the degree centrality calculated on the initial network is often much less harmful than a strategy based on the centrality recalculated at each step of percolation.

4.2 Outcome variables

The primary outcome variables are the relative size of the largest connected component reordered over after each removal step in percolation and the percolation threshold, defined as the fraction of removed nodes at which the giant component collapses. Moreover, also the average path length at each removal step will be recorded, as done by Holme et al. (2002).

4.3 Implementation

All experiments will be implemented in Python using the NetworkX library for graph manipulation, the powerlaw library for degree-distribution fitting, and Matplotlib for figures.

5. Answering the research question and feasibility of the project

The proposed design directly addresses the research question. The contrast between random and targeted removal on each real network demonstrates the "robust yet fragile" claim and provides a quantitative comparison between the Internet AS-level topology and the WWW hyperlink network. The ER and BA nulls serve as theoretical baselines: agreement with the BA prediction supports the standard scale-free explanation, while systematic deviations suggest the importance of higher-order features, beyond the degree distribution, in networks' robustness.

All analyses can be conducted on a standard laptop. Moreover, the project is feasible within the time available: we will start our analysis by using the basic model described in the previous section, and, depending on the remaining time, we will implement some of the suggested model extensions.

7. Literature

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8. [Optional] Draft timeline outlining the main steps of our project

Step 1. We acquire the datasets, the AS-level Internet topology and the WWW hyperlink network, from SNAP and conduct an initial exploratory analysis (i.e. plotting degree distributions and calculating basic network properties such as average degree, clustering coefficient).

Step 2. We implement the percolation process for both random attacks and targeted attacks based on degree centrality; conduct percolation on the AS-level Internet and the WWW network separately and compare the percolation curve and threshold between random and targeted attack for each network.

Step 3. We generate the corresponding ER and BA models for each network; we conduct percolation and compare the percolation curves with the two real world networks.

Step 4. We compare the deviations of the two real world networks from their corresponding BA model.

Step 5. Depending on the remaining time available, we implement some of the model extensions suggested previously. Ideally, although probably not feasible within this timeframe, by adding more levels of complexity, this analysis could lead to the identification of the structural features which drive the observed resilience differences (if any) between the AS-level Internet topology and the WWW hyperlink network.

Step 6. We interpret the results, discuss implications for cybersecurity strategies, limitations of our models and future work.